Evaluating Prompting Strategies in PMP-Style Certification Questions with GPT

Pedro Ivo Leite[1], Danyllo Albuquerque[1,2], João Gabriel Salvador Paiva[2], Mirko Perkusich[2], and Angelo Perkusich[2]
[1]*Federal Institute of Paraiba, LIS Group, Campina Grande, PB, Brazil*
[2]*VIRTUS Research, Development and Innovation Center, ISE Group, Campina Grande, PB, Brazil*
Corresponding author: danyllo.albuquerque@virtus.ufcg.edu.br

***Abstract*—Large Language Models (LLMs) are well-suited for text-based certification tasks due to their ability to capture patterns from large-scale language data. This study evaluates the impact of prompting strategies on performance in PMP Version 7 (PMP 7) single-best-answer questions. Using GPT-5.4-mini, zero-shot and Chain-of-Thought (CoT) prompting are compared on a dataset of 5,000 questions across multiple domains (Business, People, Process), difficulty levels (Easy, Medium, Hard), and project approaches (Agile, Hybrid, Predictive). Both strategies achieve consistently high accuracy above 97%, with CoT providing only modest gains (+0.72 percentage points overall). CoT reduces errors (from 137 to 101) and enhances performance in specific contexts, such as the Business domain and Hybrid or Predictive approaches. Both prompting strategies agree on 95.5% of the answers, demonstrating a strong baseline capability. These results indicate that modern LLMs have internalized much of the reasoning previously elicited through advanced prompting, thereby reducing the marginal benefit of more complex techniques. Overall, simpler prompting strategies are increasingly sufficient to achieve high performance, underscoring the growing maturity and robustness of contemporary LLMs.**

***Index Terms*—PMP, Project Management, Large Language Models, Prompt Engineering, Empirical Evaluation.**

I. Introduction

Large Language Models (LLMs) are increasingly recognized as tools that can support various project management activities, including planning, decision-making, risk analysis, training, and certification preparation [1]. Project management is a critical discipline for software-intensive and engineering projects, as failures are frequently linked to inadequate planning, poor communication, ineffective change management, and insufficient risk control [2]. Within this context, LLMs provide a promising form of on-demand assistance by answering questions, explaining concepts, and aiding practitioners in interpreting project management scenarios.

An expanding body of research has examined whether LLMs can demonstrate project management competence by responding to certification-style questions. Previous studies have evaluated models such as GPT-3.5, GPT-4, and Google Bard using PMP-aligned assessments, indicating that these models can achieve competitive performance in domains related to people, process, and business environment [2], [3]. However, these studies also identified significant limitations, including inconsistent answer formats, model refusals, potential data contamination, and challenges with questions requiring visual interpretation or broader contextual judgment [2], [3]. Despite these limitations, the findings suggest that LLMs may serve as complementary assistants to project managers rather than substitutes for human expertise.

However, the rapid evolution of LLMs raises a new question. Earlier work often emphasized the need for advanced prompting strategies, such as Chain-of-Thought (CoT), Faithful CoT, Tree-of-Thought, or few-shot learning, to improve model performance [3]. More recent models may have internalized part of this reasoning capability, reducing the marginal benefit of complex prompt engineering. In other words, the relevant question is no longer only whether LLMs can answer PMP-style questions, but whether advanced prompting still produces meaningful gains when applied to state-of-the-art models.

This study revisits the problem by evaluating GPT-5.4-mini on a large dataset and comparing two prompting strategies: zero-shot and chain-of-thought (CoT). The dataset consists of PMP Version 7 (PMP 7) single-best-answer multiple-choice items, which are predominantly scenario-based and emphasize decision-making within project contexts. Instead of simple recall, the questions require selecting the most appropriate action consistent with project management principles. All items were administered under controlled conditions to facilitate direct comparison between prompting strategies.

The results indicate that both prompting strategies achieve very high accuracy, exceeding 97%, which demonstrates significant advancement compared to earlier generations of large language models (LLMs). The CoT strategy further enhances performance, increasing overall accuracy from 97.26% to 97.98% and reducing the number of incorrect answers from 137 to 101. However, this improvement is modest and varies by context. Notably, improvements are more pronounced in the Business domain and in Hybrid and Predictive approaches, while gains are smaller or even negative in Agile scenarios. Additionally, both prompts yield identical answers for 95.5% of the questions, suggesting that simple prompting is sufficient for the majority of cases.

This paper makes three primary contributions. First, it presents an updated empirical assessment of modern LLM performance on project management certification-style tasks. Second, it compares zero-shot and CoT prompting to determine whether explicit reasoning continues to improve accuracy in high-performing models. Third, it analyzes performance across domains, difficulty levels, and project approaches, demonstrating that the benefits of prompting are not uniform and that modern LLMs are increasingly effective even with simple prompting strategies.

The remainder of this paper is organized as follows. Section II reviews background concepts and related work on LLMs and AI-assisted Agile practices. Section III describes the research design, dataset, and evaluation setup. Section IV presents and discusses the main findings. Section V outlines key implications. Finally, Section VI discusses threats to validity, and Section VII concludes the paper with final remarks and future directions.

II. Fundamentals and Related Work

This section reviews the project management certification ecosystem, the use of LLMs in educational and certification contexts, and prior empirical studies evaluating LLMs in PMP-style assessments. It concludes by identifying the research gap that motivates this work.

PMI and PMP Certification. Project management is a critical discipline for the successful execution of projects, encompassing planning, coordination, risk management, and stakeholder communication [4]. Despite the availability of standardized practices, many projects still fail due to inadequate management processes and insufficient alignment among stakeholders [5]. To address these challenges, the Project Management Institute (PMI) has established widely adopted frameworks such as the *Project Management Body of Knowledge (PMBOK)*, which defines a structured set of knowledge areas and best practices [6].

Within this ecosystem, the *Project Management Professional (PMP)* certification is one of the most recognized credentials for validating practitioners' knowledge and competencies [7]. The PMP exam is composed of questions that assess understanding across domains such as People, Process, and Business Environment [5]. Although certification does not guarantee real-world performance, it serves as a standardized benchmark for evaluating knowledge and decision-making in project management [7]. This structured and normative nature makes PMP-style questions an appropriate setting for evaluating the reasoning capabilities of automated systems such as LLMs.

LLMs in Educational and Certification Training. Recent advances in generative artificial intelligence have introduced LLMs as powerful tools to support education and professional training. These models can generate explanations, quizzes, and feedback in natural language, enabling personalized and scalable learning experiences [8]. Prior studies show that LLMs can enhance engagement, support self-directed learning, and assist in structuring educational content, although human supervision remains necessary to ensure contextual adequacy and correctness [9].

In project management education, LLMs have been used to design course structures, align learning objectives with assessments, and support students in project-based learning environments [9]. For instance, LLM-assisted approaches can help generate learning materials and simulate expert guidance, while prompt engineering has been incorporated into educational settings to improve decision-making skills and practical understanding [10]. More broadly, generative AI has been shown to bridge expertise gaps by enabling less experienced individuals to perform complex tasks with higher quality, particularly in planning and analytical activities [11].

Despite these benefits, concerns remain regarding the reliability of LLM outputs. These models may generate plausible but incorrect responses and are sensitive to prompt formulation, which can significantly influence performance. As a result, evaluating LLMs in controlled, certification-like environments has become an important research direction to better understand their capabilities and limitations in high-stakes contexts.

LLMs in PMP Certification Contexts. A growing body of work has specifically investigated the performance of LLMs in PMP-style certification tasks. Vakilzadeh et al. [3] evaluated GPT-3.5, GPT-4, and Google Bard on a large set of PMP-like questions, reporting accuracies ranging from 67% to 82%. Their study also demonstrated that prompting strategies can significantly improve performance, with GPT-4 reaching over 92% accuracy when such techniques are applied. Similarly, Karnouskos [2] conducted an empirical evaluation using PMP exam preparation datasets and found that all tested models achieved scores above 82%, indicating that LLMs can effectively answer project management certification questions. The study emphasizes that LLMs should be viewed as complementary assistants to project managers, enhancing access to knowledge and supporting decision-making rather than replacing human expertise.

Together, these studies provide evidence that LLMs are capable of achieving competitive performance in structured, certification-style tasks. However, they also highlight important limitations, including sensitivity to prompt design, variability across models, and challenges in handling more complex or context-dependent questions.

Research Gap and Motivation. Despite these advances, prior work has primarily focused on comparing different LLMs or demonstrating the benefits of advanced prompting strategies. Less attention has been given to how modern, high-performing LLMs behave under simpler prompting conditions, and whether the gains from complex techniques such as CoT remain substantial as models evolve.

This gap is particularly relevant given the rapid improvement of LLM capabilities. Recent models may have internalized reasoning patterns that previously required explicit prompting, reducing the marginal benefit of more sophisticated prompt engineering. Consequently, it is unclear whether complex prompting strategies are still necessary to achieve high performance in structured domains such as PMP certification tasks.

To address this gap, this study provides an updated empirical evaluation of LLM performance using PMP Version 7 (PMP 7) single-best-answer multiple-choice questions, focusing on the comparison between a zero-shot baseline and a CoT prompting strategy. By analyzing performance across domains, difficulty levels, and project approaches, we aim to understand whether prompt engineering continues to play a significant role, or whether simpler prompting strategies are increasingly sufficient for modern LLMs.

III. Research Design

This section presents the study's methodological design, including the research goal, guiding questions, and the empirical process used to investigate how prompt techniques influence LLM accuracy on PMP-style certification questions.

A. Research Goal and Questions

Guided by the Goal-Question-Metric (GQM) paradigm, our goal is: *analyze GPT, in the context of PMP-style certification questions, for the purpose of evaluating how different prompt techniques affect its factual accuracy, domain-wise performance, and error behavior, from the perspective of project management certification and training*. From this goal, we derive the following research questions (RQs):

Research Questions

RQ1. How does GPT accuracy vary between zero-shot and CoT prompting?

RQ2. How do prompt techniques affect performance across domains, difficulty levels, and project approaches?

RQ3. To what extent do prompt techniques reduce errors?

Together, these RQs provide a structured view of how prompt techniques influence LLM behavior. RQ1 focuses on quantifying overall performance differences, assessing whether more complex prompts still provide gains over a strong zero-shot baseline. RQ2 examines the consistency of these gains across contexts, such as domains, difficulty levels, and project approaches, given that improvements may not be uniform. RQ3 addresses reliability, analyzing whether prompting reduces errors and disagreement cases. This is particularly relevant in high-accuracy settings, where improvements are expected to be incremental and reflected more in stability than in large accuracy gains.

B. Empirical Process

We follow a quantitative question-response-evaluation protocol inspired by prior LLM assessment studies in certification and structured reasoning contexts [2], [3]. All experiments were conducted using GPT-5.4-mini. Focusing on a single model allows us to isolate the effect of prompt formulation without introducing confounding variability from different model architectures. The study was organized into four stages, illustrated in Figure 1.

Stage 1: Dataset Preparation. The dataset consists of 5,000 PMP-style certification questions covering three knowledge domains: *Business*, *People*, and *Process*. Questions are structured as multiple-choice items with predefined answer options and gold-standard answers. In addition, each question is annotated with difficulty level (easy, medium, hard) and project approach (Agile, Hybrid, Predictive), enabling fine-grained analysis across contexts.

All questions were stored in a structured format, including fields such as `question`, `options`, `gold_answer`, `domain`, `difficulty`, and `approach`. The dataset was validated to ensure consistency of answer keys and coverage across categories.

Stage 2: Prompt Execution. Each question was evaluated independently, without conversational memory or context carry-over, using the GPT API. We compare two prompting strategies:

Zero-shot: a direct instruction containing only the question and answer options, serving as the baseline; *Chain-of-Thought (CoT)*: an instruction that encourages the model to reason before selecting the final answer.

All executions were automated through a Python-based pipeline that iterates over the dataset, applies both prompts to each question, and stores outputs in structured JSON format. This setup ensures reproducibility and prevents cross-contamination between prompt conditions.

Stage 3: Answer Evaluation and Data Handling. Model responses were normalized and validated prior to analysis. Outputs were parsed to extract selected options, mapping textual responses into standardized answer representations. For multi-option answers, predictions were treated as unordered sets.

Accuracy was computed as the proportion of correctly predicted answers relative to the gold standard. In addition to accuracy, we tracked incorrect responses and inconsistencies between prompts. Invalid or malformed outputs were flagged and excluded from analysis when necessary.

Stage 4: Comparative Analysis. We performed a comparative analysis of the two prompting strategies across multiple dimensions. First, we computed overall accuracy and compared gains between zero-shot and CoT. Second, we analyzed performance by domain, difficulty level, and project approach to identify context-dependent variations. Third, we examined error distributions and disagreement cases, including instances where prompts produced different answers.

Finally, we conducted a qualitative inspection of error cases to identify recurring patterns and potential reasoning limitations. The results of this analysis are presented in Section IV.

IV. Results and Discussion

This section presents the empirical results obtained in April 2026 from answering 5,000 PMP certification-style questions using two prompt techniques: zero-shot and CoT. Following the GQM plan defined in Section III, we organize the results around the three research questions. Extended tables and raw analyses are available in the supplementary material (see Artifacts Availability Section).

A. Overall accuracy across prompt strategies (RQ1)

To answer RQ1, we compare the overall accuracy of GPT-5.4-min under zero-shot and CoT prompting. Both strategies achieve very high performance, exceeding 97% accuracy, confirming a strong baseline capability even under simple prompting conditions. The zero-shot approach correctly answers 4,863 out of 5,000 questions (97.26%), while CoT improves this result to 4,899 correct answers (97.98%), corresponding to a gain of +0.72 percentage points.

In absolute terms, CoT reduces the number of errors from 137 to 101, correcting 36 additional questions, which represents a 26.3% reduction in errors. This indicates that CoT primarily improves performance by addressing a subset

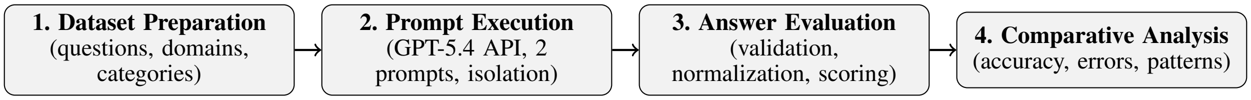

Fig. 1. Overview of the empirical process.

of previously incorrect cases rather than shifting the overall distribution of predictions. This interpretation is reinforced by the high agreement between prompts, with identical answers in 95.5% of the questions, suggesting that most responses are stable and unaffected by the prompting strategy.

Furthermore, the relatively small magnitude of improvement, combined with the already high baseline, suggests diminishing returns from more complex prompting strategies when applied to mature LLMs. In this context, CoT acts as a selective correction mechanism, helping the model resolve borderline or ambiguous cases, rather than broadly enhancing reasoning across all inputs.

Overall, the results indicate that GPT-5.4-mini already internalizes much of the reasoning required for PMP-style questions, and that prompt engineering, while still beneficial, provides incremental and targeted gains rather than substantial performance improvements.

Answer to RQ1

GPT-5.4-mini achieves high accuracy under both prompting strategies (¿97%). CoT improves performance from 97.26% (4,863/5,000) to 97.98% (4,899/5,000), correcting 36 additional questions and reducing errors by 26.3%. Overall, gains are modest and concentrated on edge cases, indicating that prompt engineering provides incremental improvements over an already strong zero-shot baseline.

B. Accuracy across domains, difficulty levels, and project approaches (RQ2)

To answer RQ2, we analyze how the impact of prompting varies across domains, difficulty levels, and project approaches, as shown in Figure 2. Each value represents the accuracy over all questions in the respective category. The results reveal that the effectiveness of CoT prompting is not uniform, but strongly dependent on the contextual characteristics of the questions.

At the domain level, CoT yields its largest improvement in the Business domain, increasing accuracy from 96.0% to 97.6% (+1.6 pp). In contrast, gains in People (+0.3 pp) and Process (+0.3 pp) are considerably smaller. This suggests that CoT is more beneficial in domains that involve interpretive reasoning or contextual judgment, while domains grounded in more structured or procedural knowledge are less sensitive to prompting variations.

A similar pattern is observed across project approaches. CoT significantly improves performance in Hybrid (97.1% → 98.4%, +1.3 pp) and Predictive contexts (96.8% → 97.9%, +1.1 pp), while a slight decrease is observed in Agile (97.9% → 97.6%, -0.3 pp). This result indicates that Agile-related questions, which tend to be more normative and principle-based, are already well captured by the model under zero-shot conditions, leaving limited room for improvement and even introducing minor instability when additional reasoning is enforced.

Across difficulty levels, CoT consistently outperforms the baseline, but the gains remain modest, ranging from +0.5 pp (hard) to +0.9 pp (easy). This relatively uniform improvement suggests that CoT provides incremental benefits regardless of difficulty, although it does not substantially alter performance in more complex cases.

Overall, these findings indicate that the benefits of prompt engineering are context-dependent and unevenly distributed. CoT is most effective in scenarios that require interpretive or integrative reasoning, while offering limited or negligible gains in well-defined, rule-based contexts. This reinforces the view that prompt techniques interact with the nature of the task, rather than providing uniform improvements across all conditions.

Answer to RQ2

The impact of prompting is context-dependent. CoT yields larger gains in Business (+1.6 pp) and in Hybrid (+1.3 pp) and Predictive (+1.1 pp) approaches, while improvements in People and Process are modest (+0.3 pp), and a slight decrease is observed in Agile (-0.3 pp). Across difficulty levels, gains are small but consistent (+0.5 to +0.9 pp). Overall, prompt effectiveness varies by context, with benefits in interpretive scenarios and limited impact in rule-based settings.

C. Error pattern across domains, difficulty levels, and project approaches (RQ3)

To answer RQ3, we analyze the distribution of incorrect answers across domains, project approaches, and difficulty levels for both prompting strategies, as illustrated in Table I. The results show that CoT primarily impacts performance by reducing errors, although this effect is uneven across contexts.

TABLE I
ERROR ACROSS PROMPTS BY DOMAIN, APPROACH, AND DIFFICULTY.

Category	Group	Baseline	PMP-Aware	Δ
Domain	Business	65	39	-26
	People	54	49	-5
	Process	18	13	-5
Approach	Agile	34	40	+6
	Hybrid	49	26	-23
	Predictive	54	35	-19
Difficulty	Easy	49	34	-15
	Medium	44	31	-13
	Hard	44	36	-8

At the domain level, CoT leads to a substantial reduction of errors in Business (65 → 39, -26 errors) and Process (18 → 13, -5 errors), while a slight decrease is observed in People (54 →

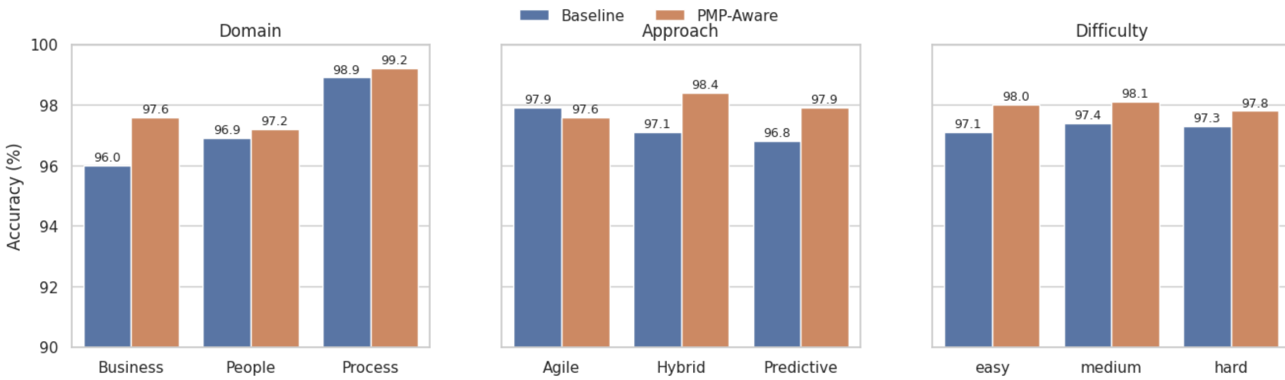

Fig. 2. Accuracy of zero-shot (Baseline) vs. CoT (PMP-Aware) across domains, approaches, and difficulty.

49 when reordered in the figure, but still lower than Business). This indicates that error reduction is concentrated in domains that require more contextual interpretation, whereas domains involving interpersonal or situational judgment remain comparatively challenging.

Across project approaches, the pattern is more heterogeneous. CoT reduces errors in Predictive (54 → 35, -19) and Hybrid (49 → 26, -23) contexts, but increases errors in Agile (34 → 40, +6). This reinforces earlier findings that Agile-related questions, which rely on strict adherence to normative principles, are already well handled by zero-shot prompting and may become less stable when additional reasoning is introduced.

In terms of difficulty, CoT consistently reduces errors across all levels, with improvements from 49 to 34 (easy), 44 to 31 (medium), and 44 to 36 (hard). Although the reductions are consistent, they are more pronounced in easier and medium questions, suggesting that CoT is more effective at correcting borderline mistakes than resolving fundamentally complex cases.

Overall, these findings indicate that CoT functions as a targeted error correction mechanism, reducing mistakes in specific contexts rather than uniformly improving performance. Importantly, the presence of error increases in certain scenarios (e.g., Agile) highlights that more elaborate prompting can introduce instability when the task requires strict adherence to well-defined rules. This reinforces the notion that prompt effectiveness depends not only on the model but also on the nature of the task and the type of reasoning required.

Answer to RQ3

CoT reduces errors overall (137 → 101), with notable gains in Business (-26), Hybrid (-23), and Predictive (-19) contexts. However, errors increase in Agile (+6), indicating instability in rule-based scenarios. Across difficulty levels, error reduction is consistent but modest. Overall, CoT acts as a targeted error correction mechanism rather than uniformly improving performance.

V. Implications

This study has implications for project management education, certification practices, and future research on LLM-based reasoning in structured knowledge domains.

For project management education and training, the consistently high accuracy observed (¿97%) suggests that modern LLMs can serve as effective tools for supporting PMP exam preparation and conceptual understanding. Their strong zero-shot performance indicates that learners can benefit even from simple interactions, without requiring complex prompt design. However, the persistence of systematic errors—particularly in domains such as Business and in specific project approaches—indicates that these tools should complement, rather than replace, guided instruction. Their educational value depends on fostering critical evaluation, prompt literacy, and awareness of potential reasoning inconsistencies.

For certification and professional practice, the results show that prompt engineering provides incremental but meaningful improvements, mainly through error reduction rather than large accuracy gains. CoT corrected 36 additional questions and reduced errors by over 26%, but also introduced instability in some contexts (e.g., Agile scenarios). This suggests that structured prompting can enhance reasoning fidelity but is not uniformly beneficial. Given that single-best-answer questions are relatively easy for modern LLMs, certification bodies should consider evolving question formats toward more complex and interpretive types. Certification providers may still use LLMs to generate questions, explanations, and feedback, provided domain experts validate accuracy. The strong performance of LLMs also reinforces the need to consider implications for assessment integrity in AI-assisted settings.

From a research perspective, PMP-style questions provide a controlled yet diverse benchmark for evaluating LLM performance across domains, difficulty levels, and project approaches. Our findings show that prompting benefits are context-dependent, with larger gains in Business, Hybrid, and Predictive settings, and limited or negative effects in Agile contexts. This highlights the importance of analyzing LLM behavior beyond aggregate accuracy. Future work should extend this analysis to other domains and investigate factors such as prompt robustness, stability, and error typologies. Overall, the results suggest that prompt engineering now plays a refinement role, improving performance at the margins while core reasoning is largely internalized, supporting more reliable human–AI collaboration in project management contexts.

VI. Threats to Validity

As with any empirical study involving LLMs, this study faces threats that should be acknowledged. Regarding **construct validity**, we operationalized "accuracy" as agreement between model outputs and predefined correct answers. In this study, all questions are "single-best answer", which reduces ambiguity and makes accuracy an appropriate and reliable metric for evaluation. Unlike settings involving multi-select or interpretive responses, this formulation minimizes partial correctness and subjective interpretation. Nevertheless, this measure does not fully capture aspects such as reasoning quality, justification adequacy, or the correctness of intermediate steps, which remain outside the scope of purely outcome-based evaluation.

Regarding **internal validity**, all prompts were executed independently, with conversation memory disabled to avoid cross-contamination. Stochastic effects were mitigated using fixed inference parameters and a controlled pipeline. The correctness of gold-standard answers is another potential threat; although no dataset is error-free, the questions were derived from PMP-style materials and organized for consistency. Data contamination is also possible, as questions may be semantically close to widely available PMP content. We mitigate this by focusing on comparative evaluation under identical conditions, ensuring equal impact across prompts.

Regarding **external validity**, the dataset consists of PMP-style certification questions covering domains, difficulty levels, and project approaches. While this provides a diverse and structured evaluation setting, generalization to other certification contexts, domains, or real-world project management scenarios may be limited. Additionally, results are specific to GPT-5.4-mini under a fixed configuration; other models, fine-tuned variants, or future versions may exhibit different behaviors.

Regarding **conclusion validity**, our findings are based on a single model and a specific evaluation setup involving two prompting strategies. Although the dataset is large (5,000 questions) and spans multiple dimensions, the observed effects—particularly the modest gains from CoT should be interpreted with caution. Prior work shows that LLM performance varies across tasks and domains, and further studies are needed to confirm whether these findings generalize to other settings, prompting techniques, and evaluation protocols.

VII. Conclusion

This paper presents an empirical evaluation of how prompting strategies influence LLM accuracy on PMP-style certification questions. Based on a dataset of 5,000 validated multiple-choice items aligned with the PMP 7 exam structure, covering project management domains, difficulty levels, and approaches, we compared two prompting techniques (i.e., zero-shot and CoT) to assess accuracy, stability, and error behavior.

The results show that both prompting strategies achieve very high accuracy (above 97%), indicating that GPT-5.4-mini would meet certification-level performance regardless of prompt choice (RQ1). CoT provides a modest improvement, increasing accuracy from 97.26% (4,863/5,000) to 97.98% (4,899/5,000) and reducing errors by 26.3%. Regarding performance across contexts (RQ2), gains are not uniform: improvements are more pronounced in the Business domain (+1.6 pp) and in Hybrid and Predictive approaches, while smaller gains are observed in People and Process, and a slight decrease occurs in Agile scenarios. Across difficulty levels, improvements are consistent but limited, indicating incremental rather than substantial gains. The analysis of error patterns (RQ3) shows that CoT acts as a targeted error correction mechanism, reducing mistakes in several contexts but introducing instability in specific cases, such as Agile questions. Overall, these findings suggest that modern LLMs already internalize much of the required reasoning, with prompt engineering mainly refining predictions in edge cases rather than broadly improving accuracy.

Future work should extend this evaluation to the upcoming PMP 8 framework, expected in July 2026, which introduces changes in exam structure and question formats. These changes are likely to involve more complex, scenario-based, and potentially adaptive question types, requiring different reasoning strategies from LLMs. Evaluating model performance under these new conditions will be essential to understanding how prompting techniques generalize to evolving certification standards. Additional directions include robustness analyses under prompt variation, multi-run stability evaluation, and deeper investigation of error typologies to better understand the limits of LLM reasoning in structured knowledge domains.

Artifacts Availability

Supplementary materials are available at: https://github.com/Ilusinusmate/softcomLLMPMP2026.

References

[1] M. B. Jelodar, "Generative ai, large language models, and chatgpt in construction education, training, and practice," *Buildings*, vol. 15, no. 6, p. 933, 2025.

[2] S. Karnouskos, "The relevance of large language models for project management," *IEEE Open Journal of the Industrial Electronics Society*, vol. 5, pp. 758–770, 2024.

[3] S. A. Vakilzadeh, S. P. Ghalejoogh, and M. Hatami, "Evaluating the potential of large language model ai as project management assistants: A comparative simulation to evaluate gpt-3.5, gpt-4, and google-bard ability to pass the pmi's pmp test," *SSRN Electronic Journal*, 2023. [Online]. Available: https://ssrn.com/abstract=4568800

[4] H. Kerzner, *Project Management: A Systems Approach to Planning, Scheduling, and Controlling*, 12th ed. John Wiley & Sons, 2017.

[5] G. G. Angel, *PMP Certification: A beginner's guide*. University Press of Colorado, 2023.

[6] Project Management Institute, "A guide to the project management body of knowledge (pmbok guide)," 2021, iSBN 978-1628256642.

[7] T. Özsoy and K. Sezgili, "Exploring the current practices and future directions in project management education and training," *Sage Open*, vol. 14, no. 1, p. 21582440241236053, 2024.

[8] I. Pesovski *et al.*, "Generative ai for customizable learning experiences," *Sustainability*, vol. 16, no. 1, 2024.

[9] E. Pereira *et al.*, "Constructive alignment in a graduate-level project management course: An innovative framework using large language models," *International Journal of Educational Technology in Higher Education*, vol. 21, no. 1, 2024.

[10] P. Mendonca *et al.*, "Using prompt engineering to enhance a project-based learning course on project management," in *Proceedings of the IEEE Frontiers in Education Conference (FIE)*, 2025.

[11] B. Hettrich *et al.*, "Bridging the expertise gap: The role of generative ai in supporting project planning tasks for novices and professionals," *Creativity and Innovation Management*, 2025.